Images in scientific writing

Images provide a pictorial insight into the research when appropriately used. Images alter the initial perception of professional editors and reviewers while assessing the manuscript.[1,2] It is essential to display images that are more relevant and explanatory to the text with a precise legend that could make the readers understand better.

Scientific images, unlike conventional photography, are data that should be of high informational value. The authors should design the figures for a wider audience that focuses on vital data with a single type of visual contrast of either color, shape, or size.[2,3] This editorial message highlights the salient features for improving the quality of images that need to be considered before submitting the manuscript to a journal.

IMAGE RESOLUTION

Image resolution is the number of pixels displayed per inch (dpi) of the image, and higher pixels provide better resolution.[4] A minimal resolution of 300 dpi is essential for submission in a scientific journal [Figure 1]. Alteration of the low-pixel images as per the journal's requirements leads to lured, soft images reducing the resolution and quality. Increasing the resolution of an image decreases the image size to compensate for the number of pixels. For example, if an image is 10" × 5" at 300 dpi, it would modify to 8" × 3" at 600 dpi. At 300 dpi, the image would fill the entire page with a good, sharp image, whereas at 600 dpi, the image size would be small, but the quality of images will be very high.[4] Photo editors potentially decrease the quality of the image: hence, it is better to capture an image at high resolution.

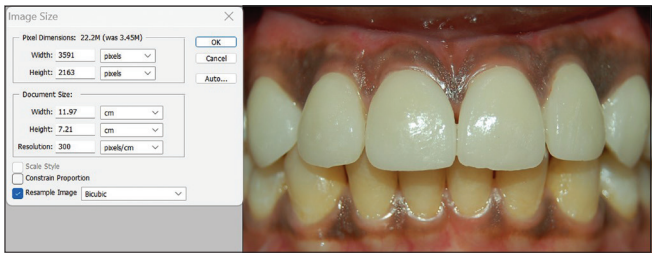

Figure 1: An image modified to 300 dpi resolution with Photo Editor software

TYPES OF IMAGE FORMATS

The quality of images depends on the type of format it is stored. The Joint Photographic Experts Group[5,6] (.jpg., jpeg) has the advantage of storing the file in small size; however, the compression distorts the image details during transfer, leading to a pixelated image. These files are easy to upload due to their small storage size. Tagged Image File Format[5,6] (.tif, tiff) is an uncompressed image file type that helps in retaining detailed and high-resolution image data even after multiple transfers of the file. The "tiff" images are versatile in the color scheme that would suit the requirement of a publication. However, the file size is larger and requires more space to upload. Portable Network Graphics[5,6] (.png) can be compressed without distortion of data. It is often used for editing image over image or text over an image. However, it does not support all color schemes. It can be used for saving line diagrams, and the file size can be small which eases the upload. The device-independent bitmap[5,6] and bitmap (.bmp) files are raster graphics that store two-dimensional images such as charts that exactly match the original source. However, the file size would be large and can be compressed using programs such as zip. Encapsulated PostScript[5,6] (.eps) is a graphic file in vector format. It is especially a master image file that can be edited and scaled to infinity without loss of resolution. It has been replaced by adobe illustrator and Portable Document Format (.pdf). The RAW image files[7] (.raw., cr2., nef., orf., srz) are raw digital negative, comprising uncompressed and unprocessed image data. The image format is usually captured by the camera sensor, and later requires software to obtain the desired output. Although the image is of high resolution, it needs specific software to read the format and requires high storage space.

TYPE OF IMAGES

The types of images that are commonly included in a dental

journal are photographs, charts, microscopic images, and radiographs.

A photographic image used in dental journals should accurately represent the color perceived by the eyes during dental/facial examination.[8] A digital single-lens reflex camera equipped with a macro lens (85–105 mm) and an external ring flash mounted in front of the lens is required for high-quality intraoral macrophotography.[9,10] The image should be captured close to the region of interest, avoiding the undesirable anatomical structures with the anti-fog mirrors. The camera is set in the manual mode for an intraoral photograph, with an aperture of f/20-22 to f/32, a shutter speed of 1/125–200, ISO 100–200, and magnification of 1:2.[9] White balance is accurately adjusted based on the neutral color of the environment so that the color is represented in its natural form depending on the light source. The white balance can be preset as sunlight, incandescent, fluorescent, etc., and can also be set based on the color temperature in kelvin. The color temperature that varies between 5500 and 6500 K gives an accurate representation of natural color to help in differentiating between healthy and diseased sites.[11] However, if the camera stores the image in RAW format, the white balance can be adjusted in postprocessing software.[12] An extraoral picture should be taken with a black background or contrast background.[13] Although most cameras prefer to store the image in jpeg format, it is preferable to store the images in the uncompressed and editable RAW format in a separate folder that can be used later to modify according to journal requirements. It is also preferable to use tiff format compared to jpeg after the final modification.

Grafts, charts, or line diagrams are vector images that are preferable to be stored in the png/tiff/pdf formats and if required to convert to jpeg format as per journal requirement.[14,15] This would ensure high resolution of the primary image during storage that can be modified when required without loss in image quality.[15] The vector images do not lose their resolution on scaling or resizing the image, however, an appropriate graphic file format such as png/tiff should be chosen. ".png" files improve the quality of graphics and are especially used while uploading to a website. Screenshots should be avoided as they may reduce the resolution of the images. The image shows saving the chart in a template (.crtx extensions) [Figure 2]. The chart can also be saved as an Excel file to enable editing of data later. The charts during submission for publication should be saved as an image file in the png/tiff/pdf formats.

Microscopic images used in publications are magnified images of an object and hence an appropriate scale of magnification

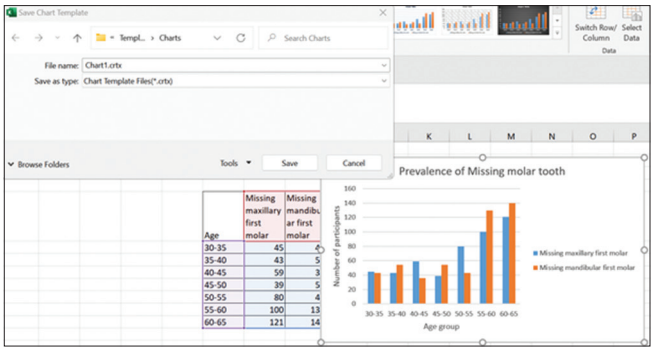

Figure 2: Chart template in .crtx extension format

is essential to satisfy the output of the research.[16,17] The scale bar is necessary for every image and should be visible in the corner of an image.[16,17] The author should make sure that the bar is maintained in the same position on all the images or follows the journal guidelines. The use of appropriate colorblind safe colors to represent the microscopical data is essential.[18] Although the image could be saved in gray or color scales, the author should perform a grayscale visibility test to ensure that the black-and-white print of color images also represents the same color variation that is visible in the color micrograph. When representing a single color, a grayscale image would be appropriate for publication. Figure 3a shows a blurred scale bar and the image due to storage of the microscopic image in jpeg format and resaving it multiple times leading to loss of data, whereas Figure 3b shows the clarity of image stored as tiff.

Radiographs are an integral part of dental treatment and should be saved in the tiff, adobe Photoshop, or Electronic Software Download to retain the highest resolution that does not lose any details.[19,20] Appropriate contrast, wide dynamic range, spatial resolution, noise reduction, and avoidance of artifact are important components of the image quality of radiographs.[20]

LEGENDS FOR FIGURES

Legends are an accurate representation of either the methodology or the results. The figure legends should be a comprehensive but detailed representation of the image. The title of the figure should be in an inactive voice which can either be a clear description of the methodology or declarative of the result from the image in a comprehensive way.[21,22] For example, a preoperative radiograph of a maxillary tooth could be better written as a preoperative intraoral periapical radiograph of the right maxillary first molar. Similar to the content of the main text, the figure should also be in the past tense. The use of symbols, colors, and scale bars should be explained in the legend. The sublegends should also be labeled as 1a and 1b to

Table 1: Preferred image format in scientific writing

Type of image	Preferred image format in storage and editing	Preferred image format during upload
Photographic image	.raw, .cr2, .nef, .orf, .srz, .tif, .tiff	.tif, .tiff, .png, .jpg, jpeg
Grafts, charts, or line diagrams	.eps, .crtx, .bmp, .png	.png, .tif, .tiff, .pdf
Microscopic or radiographic image	. tif, .tiff, .png	.tif, tiff, .png, .jpg, .jpeg

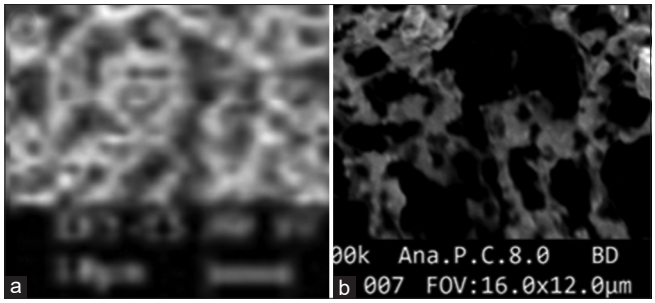

Figure 3: Scanning electron microscope image: (a) blurred image in .jpeg format after multiple formats, (b) clarity of image in .tiff format

specify the difference between the images, for example, Figure 1: scanning electron microscopy of implant surface shows (a) adherence of microbes in the coronal portion at ×100 and (b) microbial colonization in the middle third at ×500.

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An author should follow the journal guidelines for images that are more specific and vary between journals [Table 1]. The format of images given in the author guidelines in a journal should be used for uploading images.

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REFERENCES

1. Pain E. How to (seriously) Read a Scientific Paper. Science; March 16, 2016. Available from: https://www.science.org/content/article/how-seriously-read-scientific-paper. [Last accessed on 2022 Mar 07].
2. Rolandi M, Cheng K, Pérez-Kriz S. A brief guide to designing effective figures for the scientific paper. Adv Mater 2011;23:4343-6.
3. Ahmad I. Digital dental photography. Part 1: An overview. Br Dent J 2009;206:403-7.
4. University of Michigan Library. All About Images: What is Resolution?. Research Guides; February 11, 2021. Available from: https://guides.lib.umich.edu/c.php?g=282942&p=1885350. [Last accessed on 2022 Mar 08].
5. BitesizeBio. An Introduction to Digital Images in Publications; July 16, 2019. Available from: https://bitesizebio.com/43785/an-introduction-to-digital-images-in-publications/. [Last accessed on 2022 Mar 07].
6. University of Michigan Library. All About Images: Image File Formats. Research Guides; February 11, 2021. Available from: https://guides.lib.umich.edu/c.php?g=282942&p=1885348. [Last accessed on 2022 Mar 08].
7. Canon. Image Formats (RAW, JPEG, HEIF) and Compression. Available from: https://www.canon-europe.com/pro/infobank/image-file-types/. [Last accessed on 2022 Mar 07].
8. Ahmad I. Digital dental photography. Part 10: Printing, publishing and presentations. Br Dent J 2009;207:261-5.
9. Manjunath SG, Raju Ragavendra T, Sowmya K, Jayalaksshmi K. Photography in clinical dentistry – A review. Int J Dent Clin 2011;3:40-3.
10. Chandni P, Anupam S, Nitin S, Shikha G. An overview on dental photography. Int J Dent Health Sci 2016;3:581-9.
11. Raza FB, Kumar VA, Jacob A, Ali A. Assessment of a smartphone-based software application as a potential digital tool in tooth shade selection: A prospective clinical study. Quintessence Int 2021;52:608-17.
12. Bosley J. Understanding White Balance – A Beginner's Guide. Photographylife; November 06, 2020. Available from: https://photographylife.com/definition/white-balance#what-is-white-balance. [Last accessed on 2022 Mar 09].
13. Sreevatsan R, Philip K. Digital photography in general and clinical dentistry-technical aspects and accessories. Int Dent J Stud Res 2015;3:17-24.
14. Microsoft Office (n.d.) Save a Chart as a Picture. Available from: https://support.microsoft.com/en-us/office/save-a-chart-as-a-picture-254bbf9a-1ce1-459f-914a-4902e8ca9217. [Last accessed on 2022 Mar 08].
15. Wyatt A. Creating a JPG File from a Chart; June 02, 2021. Available from: https://excelribbon.tips.net/T001775_Creating_a_JPG_File_from_a_Chart.html. [Last accessed on 2022 Mar 11].
16. Jambor H, Surendranath V, Kalinka AT, Mejstrik P, Saalfeld S, Tomancak P. Systematic imaging reveals features and changing localization of mRNAs in Drosophila development. Elife 2015;4:e05003.
17. Nyakatura JA, Melo K, Horvat T, Karakasiliotis K, Allen VR, Andikfar A, *et al.* Reverse-engineering the locomotion of a stem amniote. Nature 2019;565:351-5.
18. Summerbell E. How to Make Scientific Figures Accessible to Readers with Color-Blindness. Science; September 09, 2019. Available from: https://www.ascb.org/science-news/how-to-make-scientific-figures-accessible-to-readers-with-color-blindness/. [Last accessed on 2022 Mar 15].
19. Levine D. How to obtain images from picture archiving and communication systems and ready them for publication. Radiology 2010;257:603-8.
20. Christian ME, Davidson HC, Wiggins RH 3rd, Berges G, Cannon G, Jackson G, *et al.* Digital processing of radiographic images from PACS to publishing. J Digit Imaging 2001;14:14-7.

21. Panter M. Writing an Effective Figure Legend. American Journal Experts; March, 2022. Available from: https://www.aje.com/arc/writing-effective-figure-legend/. [Last accessed on 2022 Mar 08].
22. Michelle S. Tips for Writing Outstanding Scientific Figure Legends: Scientific Writing. Bioscience Writers; June 26, 2015. Available form: https://www.biosciencewriters.com/Tips-for-Writing-Outstanding-Scientific-Figure-Legends.aspx. [Last accessed on 2022 Mar 08].

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